

Answer Key & Teacher Notes

Session 1 – Comparing Numbers on a Number Line • Session 2 – Ordering Integers and Rational Numbers

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Comparing Numbers on a Number Line

TALK PROMPTS

- How does a point's position on the number line tell you which score is greater?
- Why is the balance with the bigger digits actually the lesser number here?

WATCH FOR

- Comparing negatives by their digits ($-40 > -13$ because $40 > 13$).
- Thinking $-0.9 > -0.09$ because $0.9 > 0.09$.
- Confusing "farther from 0" with "greater."

PARTNER TASK

Number Line Pairs. One partner plots -16 and -29 on a number line from -30 to 0 ; the other writes two true inequalities, one with $<$ and one with $>$. Switch roles with the Freezer Check: -18 and 5 .

SESSION 2 • TEACHER NOTES

Ordering Integers and Rational Numbers

TALK PROMPTS

- Why can one number sit farthest from zero and still be the least number in the set?
- How does sorting the negatives apart from the positives make ordering faster?

WATCH FOR

- Ordering negatives by their digits (writing -2 , -9 , -14 as least to greatest).
- Confusing greatest absolute value with greatest number.
- Ordering mixed forms without converting them first ($-1/2$ vs -0.75).

PARTNER TASK

Card Sort. One partner orders -4.5 , 1.5 , -0.5 , and 3 from greatest to least; the other writes each absolute value. Together explain why -4.5 has the greatest absolute value but is the least number.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.NS.C.7a	1	2	1	1	2	2	2	3	3	1	1
S1 • Form B	6.NS.C.7a	1	2	1	1	2	2	2	3	3	1	1
S2 • Form A	6.NS.C.7b	1	2	1	1	2	2	2	3	3	1	1
S2 • Form B	6.NS.C.7b	1	2	1	1	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

greater than ($>$): Farther to the **right** on the number line.

less than ($<$): Farther to the **left** on the number line.

inequality: A math sentence that **compares** numbers using $<$ or $>$.

- On a number line, the number farther **right** is greater.
- For two negative numbers, the one closer to **0** is greater.

Example: -5 is to the **right** of -14 . So $-14 < -5$.

Answer: -5 is **greater**; -14 is colder.

YOU TRY

1. $a < b$.
2. $-3/4 < -1/4$

EXIT TICKET

1. $a < b$.
2. Kim; $-9 > -14$
3. **C** ($-8 < -3$)

PRACTICE FORM A

1. $a > b < c < d$.
2. a. points at $-11, -6, 0, 3$ b. $-6 > -11$ c. greatest 3 (Tomas); least -11 (Willa)
3. -32 ($-32 > -47$)
4. -18 degrees ($-18 < 5$)
5. $-24 < -9$; Cave A is deeper
6. $-1/8$ ($-7/8 < -1/8$)
7. No; -0.09 is greater ($-0.90 < -0.09$, because -0.09 is closer to 0)
8. a. $-45, -8, 12$ b. -45 c. It is farthest from 0, but it is farthest left, so it is the least balance.
9. Counting order works for positive numbers. On a number line, -15 is farther left than -4 , so -15 is less: $-15 < -4$.
10. **A** ($<$)
11. **B** (farther right)

Self-check word COMPARE

PRACTICE FORM B

1. a. $>$ b. $<$ c. $<$ d. $>$
2. a. points at $-8, -3, 0, 4$ b. $-3 > -8$ c. greatest 4 (Eli); least -8 (Maya)
3. -28 ($-28 > -53$)
4. -21 degrees ($-21 < 3$)
5. $-31 < -13$; Cave B is deeper
6. $-3/8$ ($-5/8 < -3/8$)
7. No; -0.07 is greater ($-0.70 < -0.07$, because -0.07 is closer to 0)
8. a. $-60, -9, 15$ b. -60 c. It is farthest from 0, but it is farthest left, so it is the least balance.
9. Counting order works for positive numbers. On a number line, -22 is farther left than -7 , so -22 is less: $-22 < -7$.
10. **B** ($>$)
11. **B** (-2)

Self-check word NUMBERS

GUIDED NOTES

order: To list numbers from **least** to greatest (or greatest to least).

least: The smallest number; it is farthest to the **left**.

greatest: The biggest number; it is farthest to the **right**.

- Least to greatest means **left** to right on a number line.
- Write every number in the same **form** (such as decimals) before you order.

Example: Write $\frac{3}{4}$ as a decimal: **0.75**. Negatives first: $-6 < -0.5$. Then positives: $0.75 < 1.5$. Answer: $-6, -0.5, \frac{3}{4}, 1.5$

YOU TRY

1. a. $-8, -3, 1$ b. $-2, -0.2, -0.02$
2. -20.25 m (the least number)

EXIT TICKET

1. a. $-12, -7, 0, 3$ b. $-0.3, -1/4, -0.2$
2. coldest -9 ; closest to 0: -1
3. **B** (-6.02)

PRACTICE FORM A

1. a. $-14, -9, -2$ b. $-3, -0.5, 0, 2.5$
2. a. $-7, -3, 0, 2, 5$ b. -7 ($|-7| = 7 > 5$) c. Mark (-7 is the least score)
3. $-1,105$ ft (order: $-1,105, -640, -188$)
4. $-1.25, -0.75, -1/2$; greatest $-1/2$
5. $-19, -11, -2, 0, 4$; coldest -19
6. $1/2, -2.05, -2.5, -3.4$; greatest absolute value -3.4
7. $-16.5, -16, -4 \frac{1}{2}, 0$; deepest -16.5 ft
8. a. $1.5, -2, -0.25, 0.75, -2.5$ b. $-2.5, -2, -0.25, 0.75, 1.5$ c. $-1/4$, because $|-0.25| = 0.25$ is the smallest absolute value
9. He ordered by distance from 0 as if the numbers were positive. For negatives, farther from 0 means farther left, so it is less. Correct order: $-14, -9, -2$.
10. **B** (-15)
11. **B** (-4.05)

Self-check word GREATER

PRACTICE FORM B

1. a. $-18, -11, -5$ b. $-4, -0.5, 0, 1.5$
2. a. $-6, -2, 0, 3, 4$ b. -6 ($|-6| = 6 > 4$) c. Nora (-6 is the least score)
3. $-1,320$ ft (order: $-1,320, -812, -245$)
4. $-1.5, -3/4, -0.5$; greatest -0.5
5. $-15, -8, -3, 0, 6$; coldest -15
6. $3/4, -1.05, -1.5, -2.6$; greatest absolute value -2.6
7. $-12.5, -12, -7 \frac{1}{2}, 0$; deepest -12.5 ft
8. a. $2.5, -1, -0.75, 0.5, -1.5$ b. $-1.5, -1, -0.75, 0.5, 2.5$ c. 0.5 , because $|0.5| = 0.5$ is the smallest absolute value
9. He ordered by distance from 0 as if the numbers were positive. For negatives, farther from 0 means farther left, so it is less. Correct order: $-12, -7, -3$.
10. **B** (-13)
11. **B** (-3.06)

Self-check word ORDERED

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · COMPARING NUMBERS ON A NUMBER LINE

1. $J = -4.5$, $K = -1$, $L = 3$. a. $J < K$ b. $L > K$ c. $J < L$
2. a. $<$ b. $>$ c. $<$ d. $>$ e. $>$ f. $>$
3. Points at -3.5 and -0.5 ; $-3.5 < -0.5$
4. a. $-5 < -3$ (or $-3 > -5$) b. Malik (-5 is the lower score)
5. Monday was colder: $-7 < -2$ (or $-2 > -7$), because -7 is farther left on the number line.
6. Yes. $|-6| = 6$, and -6 is left of 6 on the number line, so $-6 < 6$.

SHEET 2 · ORDERING INTEGERS AND RATIONAL NUMBERS

1. -3.25 , -2.5 , $-3/4$, 1.5 , $2\frac{1}{4}$
2. a. -9 , -6 , 0 , 4 b. -3 , -0.3 , -0.03
3. a. -2 b. -1.04 c. $-1/2$ d. 0
4. shark (-45.5), dolphin (-30.25), turtle (-12), jellyfish (-8.75)
5. He ordered by distance from 0 , as if the numbers were positive, so his list goes greatest to least. For negatives, farther from 0 is less. Correct order: -0.6 , $-1/2$, -0.4 .
6. **B** (-8 , -5 , 0 , 3)
7. **B** (-2.07)